

SATHARAKA NILMANTHA

+94 765871905 | "prasanna",Chilaw/Kudawewa | satharakanilmantha1@gmail.com | GitHub | LinkedIn | Portfolio

ABOUT ME

Junior Associate Software Developer with a strong Computer Engineering foundation, dedicated to building innovative and high-performance systems. I thrive on translating real-world problems into technical solutions, ensuring scalability and reliability at every stage of the development lifecycle. Actively engaged in expanding my technical stack to drive product innovation and operational efficiency.

EDUCATION

BSc. Engineering (Hons.) in Computer Engineering **Apr 2022 – Aug 2026**

University of Jaffna, Sri Lanka

Relevant Coursework: Software Engineering, Object-Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Computer Architecture, Embedded Systems Design, Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Advanced Networking.

G.C.E. Advanced Level – Physical Science Stream **2016–2020**

Dhammissara National Collage, Nattandiya

Results: A – Combined Mathematics, B – Chemistry, C – Physics

WORK EXPERIENCES

Software Developer - Junior Associate **May 2026- Present**

Cyber Mas Solutions (Pvt) Ltd., Colombo, Sri Lanka

VETAHR Payroll Management System (*Tech Stack:* ***Node.js, Express.js, React.js, Tailwind CSS, MySQL, Docker, Nginx, JWT, Hikvision API***)

- **Maintain** and enhance **payroll system** ensuring stability, performance, and compliance. Optimize queries and improve **fingerprint matching** via **Hikvision API** for accurate attendance.
- **Develop new features** and **RESTful APIs** with **Node.js** + **Express.js** + **JWT**. Build **responsive UI** with **React.js** + **Tailwind CSS** aligning with business needs.
- **Manage deployment** with **Docker** + **Nginx**. Collaborate with stakeholders as key **Junior Associate**, continuing development from internship.

Software Developer Intern **November 2025 – May 2026**

Cyber Mas Solutions (Pvt) Ltd., Colombo, Sri Lanka

Smart Dispatch System (*Tech Stack:* ***Java Spring Boot, React.js, Tailwind CSS, MySQL, Docker, Nginx, Grafana, JMeter, JWT***)

- **Built** microservices dispatch platform with **barcode generation/scanning** for instant package retrieval. Developed **React.js** + **Tailwind CSS** frontend consuming **REST APIs** via **Spring Cloud API Gateway**.
- **Architected Spring Boot microservices** with **Spring Cloud** for **service discovery** and **API gateway**. Implemented **JWT-based authentication** + **HTTPS security** across all services, with **MySQL** following **database-per-service isolation**.
- **Deployed** via **Docker** + **Nginx** on **Ubuntu**, integrated **Grafana** for monitoring and tested with **Apache JMeter** to validate **scalability** and **performance** under peak load.

VETAHR Payroll Management System (*Tech Stack:* ***Node.js, Express.js, React.js, Tailwind CSS, MySQL, Docker, Nginx, JWT, Hikvision API***)

- **Developed** automated **payroll system** handling attendance, leave requests, and payroll processing with **ETF/EPF contributions** and **bank salary transfers**. Integrated **fingerprint matching** via **Hikvision API** for real-time biometric attendance.
- **Built secure RESTful APIs** using **Node.js** + **Express.js** with **JWT-based authentication** and authorization. Developed **responsive UI** with **React.js** + **Tailwind CSS**, collaborating in an **Agile team** environment.
- **Deployed** stack using **Docker** + **Nginx** on cloud infrastructure with managed **cloud database**, ensuring seamless scalability and reliable performance.

▼ PROJECTS

HealthHub – Smart Hospital Management System **Jan 2025 - May 2025**

Tech Stack: ***Java Spring Boot, React.js, MySQL, Bootstrap, Postman, Jira***

- **Contributed** to full-stack development of **role-based hospital management system** using **Spring Boot** + **React.js**. Implemented **admin employee management** and **receptionist access** to doctor prescriptions.
- **Designed responsive UI** for patients, doctors, admins, and receptionists. Implemented **real-time appointment booking** and **doctor prescription features** for streamlined workflows.
- **Conducted QA testing** with manual + exploratory methods, **validated APIs** using **Postman**, and collaborated in **Agile sprints** tracked via **Jira**.

Smart Pet Feeder – IoT-Based Automated Feeding System **Jan 2025- May2025**

Tech Stack: ***ESP32, C, React Native, Firebase***

- **Developed IoT pet feeder** using **ESP32** + **C** with **ultrasonic sensors** for real-time food level monitoring and **instant alerts** via **Firebase** notifications.
- **Built React Native mobile app** for **remote control**, **feeding schedule management**, **manual override**, and **dynamic portion control** with **offline support**.
- **Utilized Firebase** for **scheduling logic**, **data storage**, and **device-user synchronization**, preventing food waste and overfeeding.

BookMyDoctor – Doctor Appointment Booking Management System **Nov 2024 - Jun 2025**

Tech Stack: ***Java Spring Boot, React.js, MySQL, Twilio, Bootstrap, bcrypt, File Handling, Chatbot Integration***

- **Developed** web-based appointment system with **secure authentication** using **bcrypt**, **file handling**, and **responsive UI** with **Bootstrap**.
- **Integrated Twilio** for **SMS notifications** including confirmations, cancellations, reminders, and **next-day alerts** for patients.
- **Built admin features** for appointment management, doctor registration/activation, and patient notifications. Integrated **React-based chatbot** for doctor details (specialization, availability, contact).

Machine Learning-Based Intrusion Detection System for Public Wi-Fi Networks **Jan 2025 - Aug 2026**

Tech Stack: ***Machine Learning, Python, Scikit-Learn, XGBoost, Pandas, NumPy, Scapy, pfSense, Mininet, Network Security***

- Developed a **real-time IDS** to detect **ARP Spoofing & MITM attacks** on public Wi-Fi using supervised ML, engineering **15 behavioral features** via **Borda Count consensus** (XGBoost Gain, Mutual Information, ANOVA, Permutation Importance).
- Deployed **SmartIDS plugin** on pfSense with **dual-layer defense** (heuristic + XGBoost), **model quantization for <1ms inference**, **ensuring 100% offline**, **privacy-first** operation with **zero internet slowdown**.
- Simulated attacks via **Mininet**, benchmarked against **Arp watch & Snort**, optimized **XGBoost pipeline** with **5-fold cross-validation** and early stopping, achieving **high accuracy with minimal false** positives for **edge deployment**.

▼ SKILLS

- **Programming Languages** : Java, Python, C++, JavaScript, HTML, CSS
- **Frameworks & Libraries** : React.js, Spring Boot, Node.js, Bootstrap, Tailwind CSS
- **Databases** : MongoDB, MySQL, Firebase
- **Tools & Technologies** : Git, GitHub, Postman, Figma, Twilio, Docker, Nginx, Grafana, JMeter
- **Network & Security** : Wireshark, WebRTC, pfSense
- **Soft Skills** : Team Collaboration, Problem Solving, Self-Learning, Leadership

▼ VOLUNTEERING EXPERIENCES

IESL Student Member **Jan 2023 – Present**

As a Student Member of the Institution of Engineers, Sri Lanka (IESL), I actively engage with the engineering community, gaining exposure to professional practices, technical knowledge, and industry standards through the institution's events and resources

Member No: ***S-31553***

▼ REFERENCES

Dr. (Mrs.) J. Jananie
Senior Lecturer Grade II,
Department of Computer Engineering,
Faculty of Engineering, University of Jaffna
jananie@eng.jfn.ac.lk
+94 70 490 4957

Mr. Sujen
Technical Lead
Cyber Mas Solutions (Pvt) Ltd,
252A Galle Road, Colombo, Sri Lanka
sujen@cybermasit.com
+94 77 554 2624